

**DATA VISUALIZATION USING POWER BI****Aim:**

To Load the data into PowerBI and transform the data and visualize it to get insights

**Algorithm:****Household Power Consumption Dashboard****Step 1: Input Data Loading**

- Read household\_power\_consumption.csv.
- Load Date, Time, Global Active Power, Voltage, Global Intensity and Sub-metering values.

**Step 2: Data Filtering**

- Check the selected date range.
- Filter the data according to the selected dates.

**Step 3: KPI Calculation**

- Calculate:
  - Total Energy Consumption
  - Average Power
  - Average Voltage
  - Maximum Power

**Step 4: Line Chart**

- Calculate average active power for each day.
- Display **Active Power vs Date** using a line chart.

**Step 5: Bar Chart**

- Calculate total values of Sub-metering 1, 2 and 3.
- Display them using a bar chart.

**Step 6: Donut Chart**

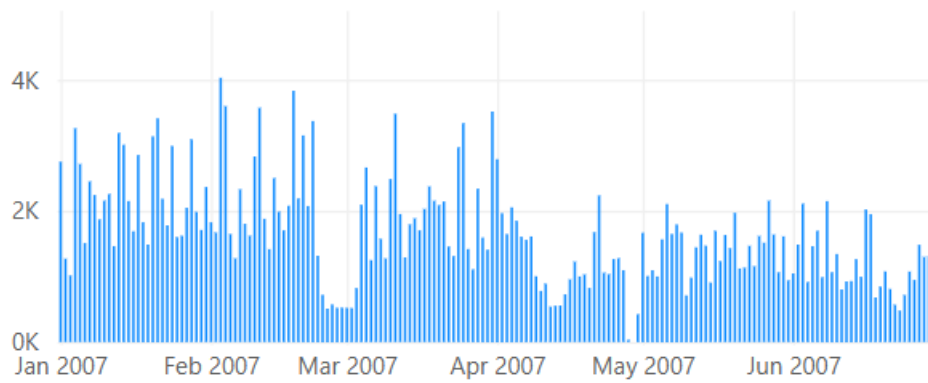
- Calculate the percentage contribution of Sub-metering 1, 2 and 3.
- Display the distribution using a donut chart.

**Step 7: Dashboard Render**

- Display KPI cards, line chart, bar chart, donut chart and date filter.
- Update all calculations when the date filter changes.

**Output:**

## Sum of Global\_active\_power by Date

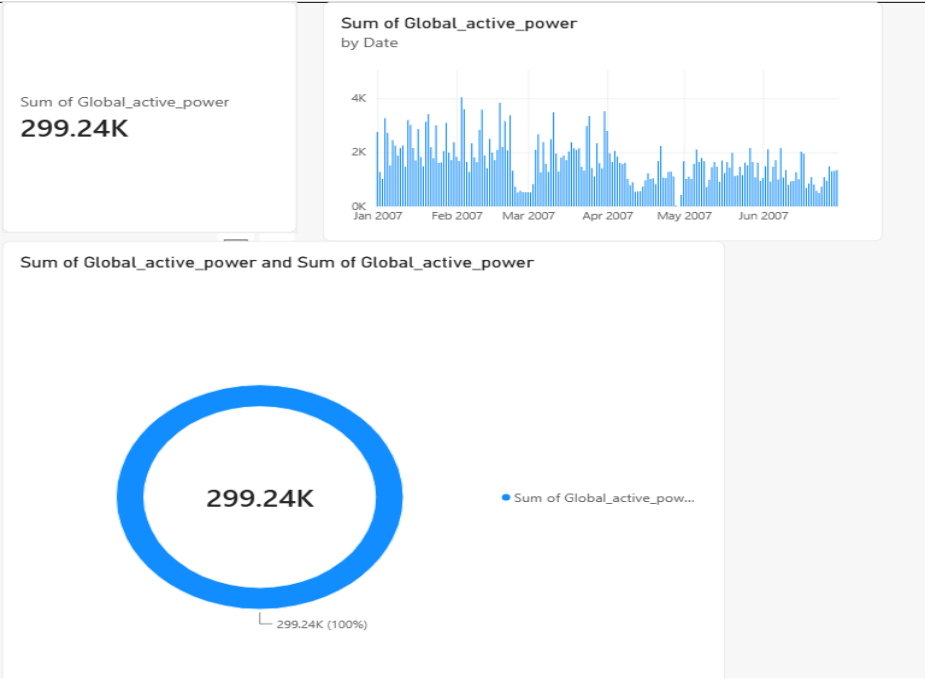
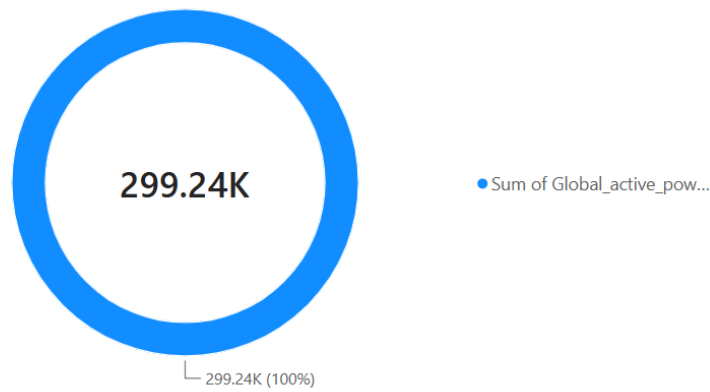


Sum of Global\_active\_power

**299.24K**

Sum of Global\_active\_power and Sum of Global\_active\_power

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**Result:** Thus the above experiment executed successfully

